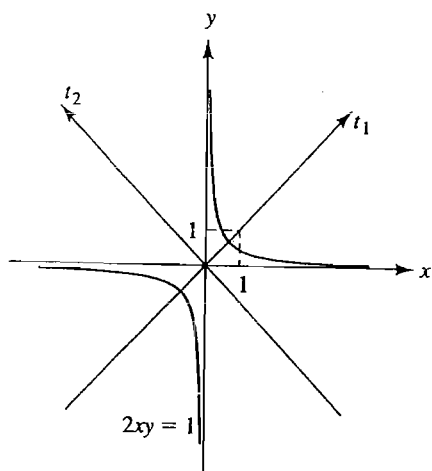


Section 8.2

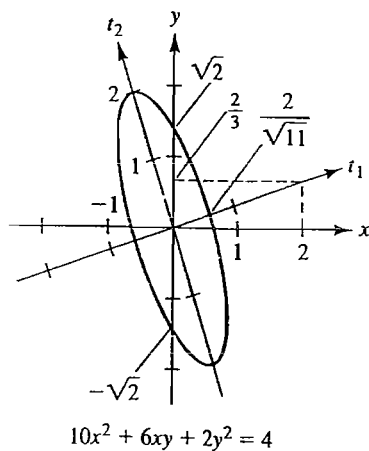
$$1. C = \begin{bmatrix} i/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$t_1^2 - t_2^2 = 1$$



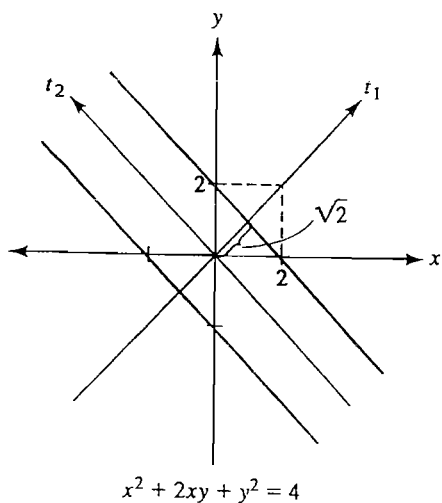
$$5. C = \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{bmatrix}$$

$$11t_1^2 + t_2^2 = 4$$



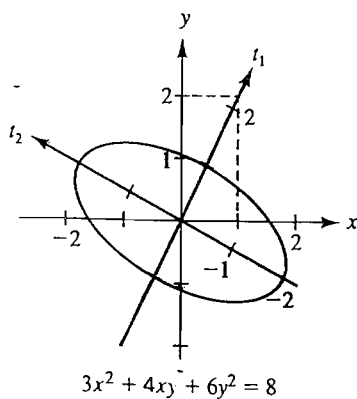
$$3. C = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

$$2t_1^2 + 0t_2^2 = 4$$



$$7. C = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

$$7t_1^2 + 2t_2^2 = 8$$



9. The symmetric matrix of the quadratic-form portion is $\begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$. Thus,

$$\det(A - \lambda I) = \begin{vmatrix} a - \lambda & b/2 \\ b/2 & c - \lambda \end{vmatrix} =$$

$$\lambda^2 - (a + c)\lambda + ac - b^2/4.$$

The eigenvalues are given by $\lambda =$

$$\left(\frac{1}{2}\right)(a + c \pm \sqrt{(a + c)^2 + b^2 - 4ac});$$
 they

are real numbers because A is symmetric; they have the same algebraic sign if $b^2 - 4ac < 0$, and they have the opposite algebraic sign if $b^2 - 4ac > 0$. One of them is zero if $b^2 - 4ac = 0$. We obtain a (possibly degenerate) ellipse, hyperbola, or parabola accordingly.

11. Let $C^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & b/r & c/r \\ 0 & -c/r & b/r \end{bmatrix}$, where

$r = \sqrt{b^2 + c^2}$. Then C is an orthogonal matrix such that $\det(C) = 1$ and

$$\begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} = \mathbf{t} = C^T \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ (by + cz)/r \\ (-cy + bz)/r \end{bmatrix}.$$

Thus,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C\mathbf{t} = \begin{bmatrix} t_1 \\ (bt_2 - ct_3)/r \\ (ct_2 + bt_3)/r \end{bmatrix}$$

represents a rotation of axes that transforms the equation $ax^2 + by + cz = d$ into the form $at_1^2 + rt_2 = d$.

13. Hyperboloid of two sheets
 15. Hyperboloid of one sheet
 17. Hyperbolic cylinder
 19. Hyperboloid of one sheet
 21. Circular cone or hyperboloid of one or two sheets
 23. Elliptic cone or hyperboloid of one or two sheets
 25. Parabolic cylinder or two parallel planes or one plane or empty
 27. Hyperbolic paraboloid or hyperbolic cylinder or two intersecting planes

Section 8.3

1. g has a local minimum of -7 at $(0, 0)$.
3. g has no local extremum at $(0, 0)$.
5. g has a local maximum of 3 at $(-5, 0)$.
7. g has no local extremum at $(0, 0)$.
9. The behavior of g at $(3, 1)$ is not determined.
11. g has a local maximum of 4 at $(0, 0, 0)$.
13. g has no local extremum at $(0, 0, 0)$.
15. g has no local extremum at $(7, -6, 0)$.
17. $g(x, y) = y^2 + 10$ 19. $g(x, y) = y^2 +$
21. $g(x, y) = x^4 + y^4 + 40$
23. The maximum is $.5$ at $\pm(1/\sqrt{2})(1, 1)$. The minimum is $-.5$ at $\pm(1/\sqrt{2})(1, -1)$.
25. The maximum is 9 at $\pm(1/\sqrt{10})(-1, 3)$. The minimum is -1 at $\pm(1/\sqrt{10})(3, 1)$.
27. The maximum is 6 at $\pm(1/\sqrt{2})(1, -1)$. The minimum is 0 at $\pm(1/\sqrt{2})(1, 1)$.
29. The maximum is 3 at $\pm(1/\sqrt{3})(1, -1, 1)$. The minimum is 0 at $\pm(1/\sqrt{2a^2 + 2b^2 - 2ab})(a - b, a, b)$.
31. The maximum is 2 at $(\pm 1/\sqrt{2a^2 + 2b^2})(a, b, -a)$. The minimum is -2 at $\pm(1/\sqrt{2})(0, -1, 1, 0)$.
33. The local maximum of f is $\lambda_1 a^2$, where λ_1 is the maximum eigenvalue of the symmetric coefficient matrix of the form f ; it is assumed at any eigenvector corresponding to λ_1 and of length a . An analogous statement holds for the local minimum of f .
35. g has a local maximum of 5 at $(0, 0, 0)$.
37. g has no local extremum at $(0, 0, 0)$.

Section 8.4

$$1. \mathbf{w}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix} 1 \\ -1/3 \end{bmatrix}, \mathbf{w}_4 = \begin{bmatrix} .75 \\ -1 \end{bmatrix}$$

Rayleigh quotients: $-2, 1, 5, 2$

Maximum eigenvalue 6 , eigenvector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$3. \mathbf{w}_2 = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix} 1 \\ 19 \\ 29 \end{bmatrix}, \mathbf{w}_4 = \begin{bmatrix} 1 \\ 65 \\ 103 \end{bmatrix}$$

Rayleigh quotients: $6, \frac{298}{74} \approx 4, \frac{4222}{1202} \approx 3.5$

Maximum eigenvalue 3, eigenvector $\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$

$$5. 5 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$7. 2 \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} - \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} + 0\mathbf{b}_3\mathbf{b}_3'$$

$$9. \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \quad 11. \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{bmatrix}$$

$$3. \lambda = 12, \mathbf{v} = [-.7059, 1, -.4118]$$

$$5. \lambda = 6, \mathbf{v} = [-.9032, 1, -.4194]$$

$$7. \lambda = .1883, \mathbf{v} = [1, .2893, .3204]$$

$$9. \lambda_1 = 4.732050807568877,$$

$$\mathbf{v}_1 = r[1, -.7320508, 1]$$

$$\lambda_2 = 1.267949192431123,$$

$$\mathbf{v}_2 = s[.3660254, 1, .3660254]$$

$$\lambda_3 = -4, \mathbf{v}_3 = t[-1, 0, 1],$$

for nonzero r, s, t

$$1. \lambda_1 = 16.87586339619508,$$

$$\mathbf{v}_1 = r[.9245289, 1, .3678643, .7858846]$$

$$\lambda_2 = -15.93189429348535,$$

$$\mathbf{v}_2 = s[-.3162426, .6635827, -1, -.004253739]$$

$$\lambda_3 = 6.347821447472841,$$

$$\mathbf{v}_3 = t[-.5527083, .9894762, .8356429, -1]$$

$$\lambda_4 = -.291790550182573,$$

$$\mathbf{v}_4 = u[-1, .06924058, .3582734, .9206089]$$

for nonzero r, s, t , and u

$$3. \text{ a. The characteristic polynomial } |A - \lambda|$$

$$= \begin{vmatrix} a - \lambda & b \\ b & c - \lambda \end{vmatrix} = \lambda^2 + (-a - c)\lambda +$$

$(ac - b^2)$ has roots

$$\lambda = \frac{1}{2} (a + c \pm \sqrt{(a + c)^2 - 4(ac - b^2)})$$

$$= \frac{1}{2} (a + c \pm \sqrt{(a - c)^2 + 4b^2}).$$

b. If we use part a, the first row vector of $A - \lambda I$ is

$$[a - \lambda, b] =$$

$$\left[\frac{1}{2} (a - c \mp \sqrt{(a - c)^2 + 4b^2}), b \right]$$

$$= [g \mp \sqrt{g^2 + b^2}, b].$$

c. From part a, eigenvectors for the matrix are $[-b, g \pm \sqrt{g^2 + b^2}] =$

$[-b, g \pm h]$. Normalizing, we obtain

$\frac{(-b, g \pm h)}{\sqrt{b^2 + (g \pm h)^2}}$. Using the upper choice of sign and setting $r = \sqrt{b^2 + (g + h)^2}$, we obtain $[-b/r, (g + h)/r]$ as the first column of C . Using the lower choice of sign and setting $s = \sqrt{b^2 + (g - h)^2}$, we obtain $[-b/s, (g - h)/s]$ as the second column of C .

$$\text{d. } \det(C) = \frac{-b(g - h)}{rs} + \frac{b(g + h)}{rs} = \frac{2bh}{rs},$$

because $h, r, s \geq 0$, we see that the algebraic sign of $\det(C)$ is the same as that of b .

$$25. \lambda_1 = -12.00517907692924,$$

$$\lambda_2 = 7.906602974286551,$$

$$\lambda_3 = 17.09857610264269$$

$$27. \lambda_1 = -5.210618568922174,$$

$$\lambda_2 = 2.856693936892428,$$

$$\lambda_3 = 3.528363748899602,$$

$$\lambda_4 = 7.825560883130143$$

$$29. 5.823349919059785,$$

$$-11.91167495952989 \pm$$

$$1.357850063519836i$$

$$31. 57.22941613544168,$$

$$-92.88108454947197,$$

$$-54.25594801085533,$$

$$47.45380821244281 \pm$$

$$44.48897425527453i$$

CHAPTER 9

Section 9.1

$$1. \text{ a. } z + w = 4 + i, zw = 5 + 5i$$

$$\text{b. } z + w = 3 + 2i, zw = -1 + 3i$$

3. a. $|z| = \sqrt{13}$, $\bar{z} = (3 - 2i)$, $z\bar{z} = (3 + 2i)(3 - 2i) = 13 = |z|^2$
 b. $|z| = \sqrt{17}$, $\bar{z} = 4 + i$, $z\bar{z} = (4 - i)(4 + i) = 17 = |z|^2$
7. a. $\frac{3}{2} + \frac{1}{2}i$ b. $\frac{13}{25} + \left(-\frac{9}{25}\right)i$
9. a. Modulus $2\sqrt{2}$, principal argument $3\pi/4$
11. 16 17. F T F F T F F T F T
19. $\sqrt{2} + \sqrt{2}i$, $-\sqrt{2} + \sqrt{2}i$, $-\sqrt{2} - \sqrt{2}i$, $\sqrt{2} - \sqrt{2}i$
21. 1, i , -1 , $-i$
23. 2, $\sqrt{2} + \sqrt{2}i$, $2i$, $-\sqrt{2} + \sqrt{2}i$, -2 , $-\sqrt{2} - \sqrt{2}i$, $-2i$, $\sqrt{2} - \sqrt{2}i$

Section 9.2

3. $AB = \begin{bmatrix} -3 + 2i & 2i & 2i \\ 2 & 2i & 1 \\ 2 + 3i & -1 + i & 2 + i \end{bmatrix}$,
 $BA = \begin{bmatrix} -2 + 2i & i & 2 - i \\ 2 + 3i & 1 + 3i & 0 \\ 2i & -1 + i & 0 \end{bmatrix}$
5. $\frac{1}{3} \begin{bmatrix} 2 + i & -i \\ -1 - i & 1 \end{bmatrix}$
7. $\frac{1}{10} \begin{bmatrix} 9 - 3i & 1 + 3i & -4 + 8i \\ -3 + i & 3 - i & -2 - 6i \\ -2 + 4i & 2 - 4i & 2 - 4i \end{bmatrix}$
9. $z = \frac{1}{10} \begin{bmatrix} -7 + 9i \\ 9 - 3i \\ 6 - 2i \end{bmatrix}$ 11. $\text{sp} \left\{ \begin{bmatrix} 1 + i \\ 1 + 3i \\ 2 \end{bmatrix} \right\}$
13. 3 15. a. $\langle u, v \rangle = 0$, $\langle v, u \rangle = 0$
 b. $\langle u, v \rangle = 5 - 3i$, $\langle v, u \rangle = 5 + 3i$
21. a. Perpendicular d. Parallel
 b. Parallel e. Perpendicular
 c. Neither
23. $\frac{2}{\sqrt{7}}[i, 1 - i, 1 + i, 1 - i]$
25. $[-3i, 1, 2 + 2i]$
27. $\{[2 + i, 1 + i], [1 - i, -2 + i]\}$
29. $\{[1, i, i], [1 + 3i, 3 - 2i, i], [1 + i, i, 1 - 2i]\}$

31. a. Both
 b. Hermitian but not unitary
 c. Not Hermitian but unitary
 d. Neither

33. T T F F T T T T F F

41. Diagonal matrices with entries of modulus 1 on the diagonal.

M1. See answer to Exercise 3.

$$\text{M3. } \begin{bmatrix} -i & 1 + i & 0 \\ 1 + i & -1 + i & 1 \\ -1 + i & -1 - 2i & i \end{bmatrix} \quad \text{M5. } \begin{bmatrix} 2 + i \\ -4 + i \\ -6i \end{bmatrix}$$

M7. 2

M9. Entering $[Q, R] = \text{qr}(A)$, where A is the matrix having the given vectors a_1, a_2, a_3 as column vectors, returns a matrix Q having as column vectors an *orthonormal* basis $\{q_1, q_2, q_3\}$, where

$$q_1 \approx [-0.5774, -0.5774i, -0.5774i],$$

$$q_2 \approx [-0.4695 - 0.1719i, -0.4695 - 0.1719i, 0.2977 + 0.6414i],$$

$$q_3 \approx [0.4695 + 0.429i, 0.0726 - 0.6414i, 0.3703 + 0.1719i]$$

To check, using MATLAB, the Student's Solutions Manual's answers

$$v_1 = a_1, v_2 = a_2, v_3 = [1 - 3i, -3 + i, -2]$$

for an *orthogonal* basis, enter

$$((1-i)/Q(1,1))*Q(:,1) \text{ to check } v_1, \text{ enter}$$

$$(1/Q(1,2))*Q(:,2) \text{ to check } v_2, \text{ and enter}$$

$$((1-3i)/Q(1,3))*Q(:,3) \text{ to check } v_3.$$

M11. a. $\sqrt{274}$, $\sqrt{476}$, $\sqrt{458}$, and $\sqrt{353}$ for rows 1, 2, 3, and 4, respectively.

b. $\sqrt{277}$, $\sqrt{192}$, $\sqrt{529}$, $\sqrt{124}$, and $\sqrt{439}$ for columns 1, 2, 3, 4, and 5, respectively.

$$\text{c. } -45 - 146i$$

$$\text{d. } 31 + 14i$$

Section 9.3

$$1. U = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$3. U = \begin{bmatrix} (1+i)/\sqrt{3} & (1+i)/\sqrt{6} \\ -1/\sqrt{3} & 2/\sqrt{6} \end{bmatrix}, D = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix}$$

$$5. U = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & 0 & i \\ i & 0 & 1 \\ 0 & \sqrt{2} & 0 \end{bmatrix}, D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$7. U = \begin{bmatrix} (1-i)/\sqrt{6} & 0 & (1-i)/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$9. U = \begin{bmatrix} (1+i)/\sqrt{6} & 0 & (1+i)/\sqrt{3} \\ 0 & 1 & 0 \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \end{bmatrix},$$

$$D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$11. U = \begin{bmatrix} (-1-i)/\sqrt{8} & 0 & (3+3i)/\sqrt{24} \\ (1-i)/\sqrt{8} & (1+i)/\sqrt{3} & (1-i)/\sqrt{24} \\ 2/\sqrt{8} & -i/\sqrt{3} & 2/\sqrt{24} \end{bmatrix},$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$13. \{a \in \mathbb{C} \mid |a| = 4\}$$

$$15. a = -1$$

$$19. \text{F T T F T F T T F F}$$

tion 9.4

1. Yes

3. No

5. No

$$7. a. \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = -2.$$

- b. $J + 2I$ has rank 3 and nullity 1,
 $(J + 2I)^2$ has rank 2 and nullity 2,
 $(J + 2I)^3$ has rank 1 and nullity 3,
 $(J + 2I)^4$ has rank 0 and nullity 4
for $k \geq 4$.

$$c. J + 2I: e_4 \rightarrow e_3 \rightarrow e_2 \rightarrow e_1 \rightarrow 0.$$

$$d. Je_1 = -2e_1, Je_2 = e_1 - 2e_2, \\ Je_3 = e_2 - 2e_3, Je_4 = e_3 - 2e_4.$$

$$9. a. \lambda_1 = -1, \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = 2.$$

$$b. (J + I)^k \text{ has rank 4 and nullity 1 for } k \geq 1.$$

$$(J - 2I) \text{ has rank 3 and nullity 2,}$$

$$(J - 2I)^k \text{ has rank 1 and nullity 4 for } k \geq 2.$$

$$c. J + I: e_1 \rightarrow 0,$$

$$J - 2I: e_3 \rightarrow e_2 \rightarrow 0, e_5 \rightarrow e_4 \rightarrow 0.$$

$$d. Je_1 = -e_1, Je_2 = 2e_2, Je_3 = e_2 + 2e_3, \\ Je_4 = 2e_4, Je_5 = e_4 + 2e_5.$$

$$11. \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$$

$$13. \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

$$15. \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \left\{ \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\} \text{ (Other bases are possible.)}$$

$$17. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 4 \end{bmatrix}, \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ (Other answers are possible.)}$$

$$19. \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}, \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

(Other answers are possible.)

$$21. \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}, \{e_1 + e_3, e_5, e_2, e_4, e_1 - e_3\}$$

(Other answers are possible.)

$$23. \text{T F T T F T T F F F} \quad 25. O \quad 27. O$$

$$29. A^4 + (3 - i)A^3 + (3 - 3i)A^2 + (1 - 3i)A - iI$$

CHAPTER 10

Section 10.1

- There are $n - 1$ flops performed on b while the first column of A is being fixed up, $n - 2$ flops while the second column of A is being fixed up, and so on. The total number is $(n - 1) + (n - 2) + \cdots + 2 + 1 = n(n - 1)/2$, which has order of magnitude $n^2/2$ for large n .
- mn , if we call each indexed addition a flop
- mn^2 7. $2n^3$ 9. $3n^3$ 11. $6n^3$
- $3n^3/2$ 15. n^3 17. $2n$
- w^2n , counting each final division as a flop
- FTFTFTFTFT
- (No text answer data are possible for this problem, since different computers run at different speeds. However, for n large enough to require 6 seconds or more to solve an $n \times n$ system, our computer did require roughly 50% more time when using the Gauss-Jordan method than when using the Gauss method with back substitution.)
- See answer to Exercise 23.

Section 10.2

- It is not significant; no arithmetic operations are involved, just storing of indexed values.

$$3. \begin{bmatrix} -18 \\ 5 \end{bmatrix} \quad 5. \begin{bmatrix} -27 \\ -11 \\ -1 \end{bmatrix}$$

$$7. \begin{bmatrix} 1 \\ -2 \\ 1 \\ 3 \end{bmatrix}$$

$$9. P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix},$$

$$U = \begin{bmatrix} 2 & 1 & -3 \\ 0 & -2 & 8 \\ 0 & 0 & 1 \end{bmatrix}, x = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$$

$$11. L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & -10 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 5 \\ 0 & 0 & 55 \end{bmatrix},$$

$$x = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$13. L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 4 & 2 & -\frac{3}{2} & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 3 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$15. \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad 17. \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$$

$$19. LDU = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & - \\ 0 & 1 & \\ 0 & 0 & \end{bmatrix}$$

$$21. FTFTFTFTFT$$

23. Display:

$$\begin{bmatrix} 1 & 3 & -5 & 2 & ; \\ 4 & -18 & 30 & 0 & -1 \\ 3 & .1666667 & 9 & -2 & 4.166 \\ 2 & .2777778 & 1.467407 & -4.185185 & 5.412 \\ -6 & -1.166667 & .6666667 & -4.141593 & 45.476 \end{bmatrix}$$

$$\text{For } b_1, x = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 2 \\ -2 \end{bmatrix}; \text{ for } b_2, x = \begin{bmatrix} -5 \\ -8 \\ 9 \\ 11 \\ 4 \end{bmatrix}; \text{ for } b_3, x = \begin{bmatrix} -5 \\ -8 \\ 9 \\ 11 \\ 4 \end{bmatrix}$$

$$25. \begin{bmatrix} .4814815 \\ -1.592593 \end{bmatrix} \quad 27. \begin{bmatrix} .8125 \\ 1.875 \\ -.3125 \end{bmatrix}$$

$$29. \begin{bmatrix} -.09131459 \\ -.08786787 \\ -.4790014 \\ -.01573263 \\ .08439226 \\ .2712737 \end{bmatrix}$$

31. The ratios roughly conform on our computer. One solution that requires 100 flops took about $\frac{1}{3}$ of the time to form the L/U display, which requires about $\frac{1000}{3}$ flops, etc.
33. The ratios roughly conform on our computer, in the sense explained in the answer to Exercise 31.

Section 10.3

1. $x_1 = 0, x_2 = 1$
3. $x_1 = 1, x_2 = .9999$. Yes, it is reasonably accurate. $10x_1 + 1000000x_2 = 1000000$, $-10x_1 + 20x_2 = 10$ is a system that can't be solved without pivoting by a five-figure computer.
5. $x_1 + 10^{19}x_2 = 10^{19}, -x_1 + 2x_2 = 1$
7. We need to show that $(n(nA)^{-1})A = I$. It is easy to see that $(rA)B = A(rB) = r(AB)$ for any scalar r and matrices A and B such that AB is defined. Thus we have $(n(nA)^{-1})A = (nA)^{-1}(nA) = I$.
9. On our computer, $[x_1, x_2] = [-10^{-9}, 10^{-9}]$, which is approximately correct.
11. MATCOMP yielded the solution $x_1 = -10^{-9}, x_2 = 10^{-9}$, which is approximately correct for the system
- $$10^{-9}x_1 + 10^9x_2 = 1$$
- $$10^9x_1 + 2(10^9)x_2 = 1.$$
13. Both with and without the scaling routine, MATCOMP successfully inverted the given matrix on our computer.
15. For $\mathbf{b}$, we have $\mathbf{x} = \begin{bmatrix} -46 \\ 96 \end{bmatrix}$, and for $\mathbf{c}$, we

$$\text{have } \mathbf{x} = \begin{bmatrix} -36 \\ 78 \end{bmatrix}.$$

The two components of $\mathbf{b}$ differ from the corresponding ones of $\mathbf{c}$ by 10 and by 18.

17. For $\mathbf{b}$, we have $\mathbf{x} = \begin{bmatrix} 32 \\ 240 \\ -1500 \\ 1400 \end{bmatrix}$, and for $\mathbf{c}$, we

$$\text{have } \mathbf{x} = \begin{bmatrix} 268 \\ -2100 \\ 3720 \\ -1820 \end{bmatrix}. \text{ Corresponding}$$

components of the solutions of $H_4\mathbf{x} = \mathbf{b}$ and of $H_4\mathbf{x} = \mathbf{c}$ differ by as much as 5220.

19. We find that $H_2^{-1} = \begin{bmatrix} 4 & -6 \\ -6 & 12 \end{bmatrix}$.

21. We were able to complete a reduction for inverses of the 1st, 2nd, 4th, 8th, and 16th powers of H_4 . The inverse checks for H_4 and H_4^2 were pretty good, and we expect that

$$H_4^{-1} = \begin{bmatrix} 16 & -120 & 240 & -140 \\ -120 & 1200 & -2700 & 1680 \\ 240 & -2700 & 6480 & -4200 \\ -140 & 1680 & -4200 & 2800 \end{bmatrix}$$

and

$$(H_4^2)^{-1} \approx \begin{bmatrix} 91856 & -1029120 & 2471040 & -1603840 \\ -1029120 & 11566800 & -27820800 & 18076800 \\ 2471040 & -27820800 & 66978000 & -43545600 \\ -1603840 & 18076800 & -43545600 & 28322000 \end{bmatrix}$$

The entries in our "inverse" of H_4^4 were of the right order of magnitude; if the entries of H_4^m are of order of magnitude roughly 10^m , then we expect the entries of H_4^{2m} to be of order roughly 10^{2m} . Also, our inverse matrix for H_4^4 was symmetric, but the inverse check was not very good. For H_4^8 and H_4^{16} , our "inverses" contained entries of completely wrong orders of magnitude, and the inverse matrices were not symmetric.

- M1. For $n = 5, \mathbf{b} \approx [1, 0]$;
 for $n = 10, \mathbf{b} \approx [1, 0]$;
 for $n = 15, \mathbf{b} \approx 10^3[1.68, -2.90]$;
 for $n = 20, \mathbf{b} \approx 10^{10}[6.265, -6.688]$;
 for $n = 25, \mathbf{b} \approx 10^{18}[2.09, -3.09]$;
 for $n = 30, \mathbf{b} \approx 10^{25}[-6.56, 6.97]$.

APPENDIX A

1. Let $P(n)$ be the equation to be proved. Clearly $P(1)$ is true, because

$$1 = \frac{1(1+1)(2+1)}{6}. \text{ Assume that } P(k) \text{ is}$$

true.

Then

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= \frac{(k+1)(2k^2 + k + 6k + 6)}{6}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}.$$

Thus, $P(k+1)$ is true, and $P(n)$ holds for all $n \in \mathbb{Z}^+$.

3. Let $P(n)$ be the equation to be proved. We see that $P(1)$ is true. Assume that $P(k)$ is true. Then

$$1 + 3 + 5 + \cdots + (2k-1) + (2k+1) \\ = k^2 + 2k + 1 = (k+1)^2,$$

as required. Therefore, $P(n)$ holds for all $n \in \mathbb{Z}^+$.

5. Let $P(n)$ be the equation to be proved. We see that $P(1)$ is true, because $a(1-r^2)/(1-r) = a(1+r) = a+ar$. Assume that $P(k)$ is true. Then

$$a + ar + ar^2 + \cdots + ar^k + ar^{k+1}$$

$$= \frac{a(1-r^{k+1})}{1-r} + ar^{k+1}$$

$$= \frac{a(1-r^{k+1} + r^{k+1}(1-r))}{1-r}$$

$$= \frac{a(1-r^{k+2})}{1-r},$$

which establishes $P(k+1)$. Therefore, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

7. The notion of an “interesting property” has not been made precise; it is not well defined. Moreover, we work in mathematics with two-valued logic: a statement is either true or false, *but not both*. The assertion that not having an interesting property would be an interesting property seems to contradict this two-valued logic. We would be saying that the integer both has and does not have an interesting property.

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